Bird Model Analysis

This notebook look into correlations and properties of the bird model.

```
======================================================================
MODULE VALIDATION CHECK
======================================================================

[1] Checking Centraliser.TensorProductGroup...
    ✓ TensorProductGroup class exists with required methods

[2] Checking Centraliser.TensorProductGroupSO (SO(N) framework)...
    ✓ TensorProductGroupSO class exists with required methods

[3] Checking Centraliser.SON (SO(N) basis)...
    ✓ SON class exists with so_basis method

[4] Checking find_generators_coefficients parameters...
    ✓ Parameters: ['self', 'M', 'verbose', 'basis_generators', 'reference_coeffs']

[5] Checking MatrixProductState.find_generators_coefficients wrapper...
    ✓ Wrapper method exists with parameters: ['self', 'M', 'basis', 'verbose', 'dim',
'reference_coeffs']
```

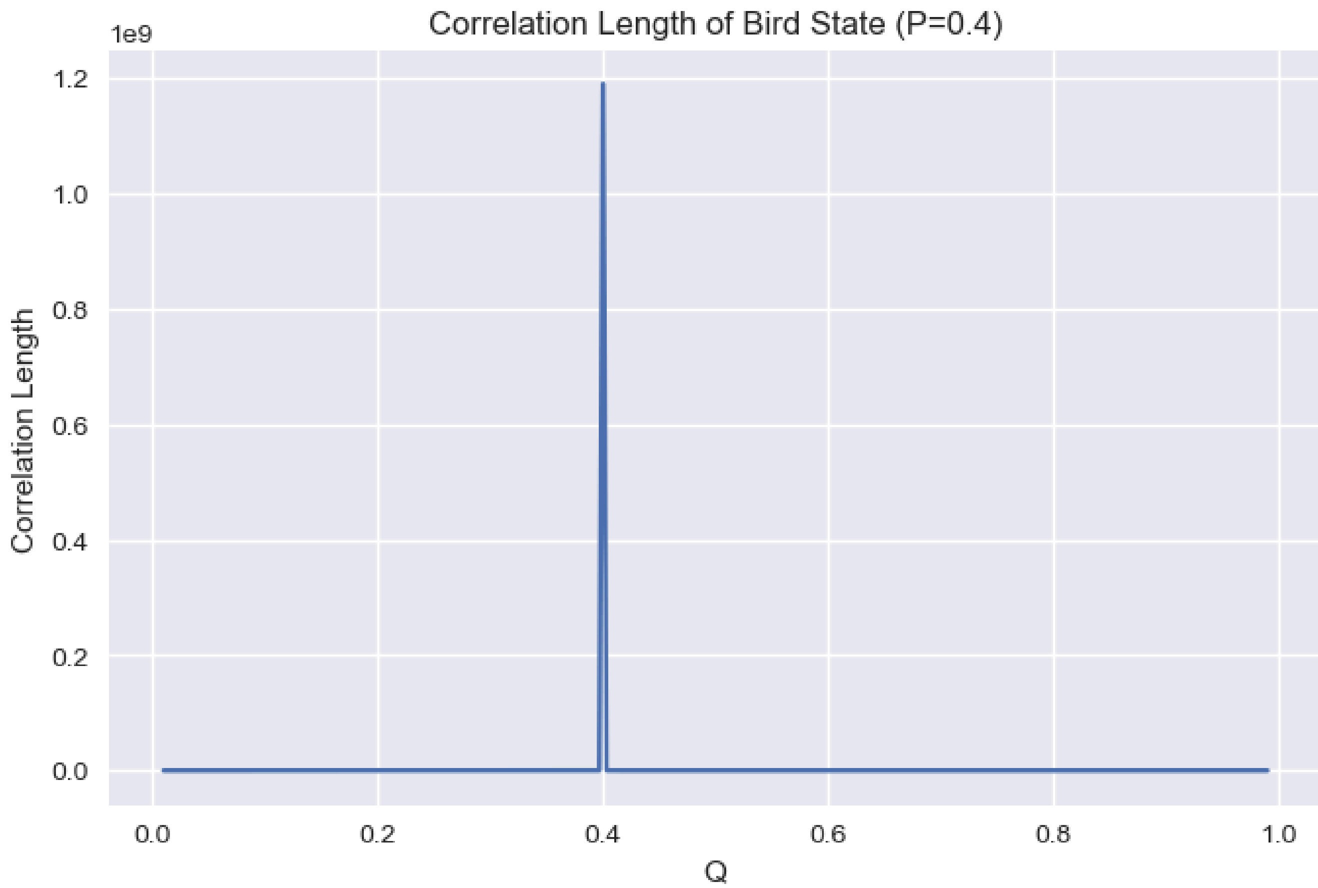
Form of the model

The site matrices are given by the form:

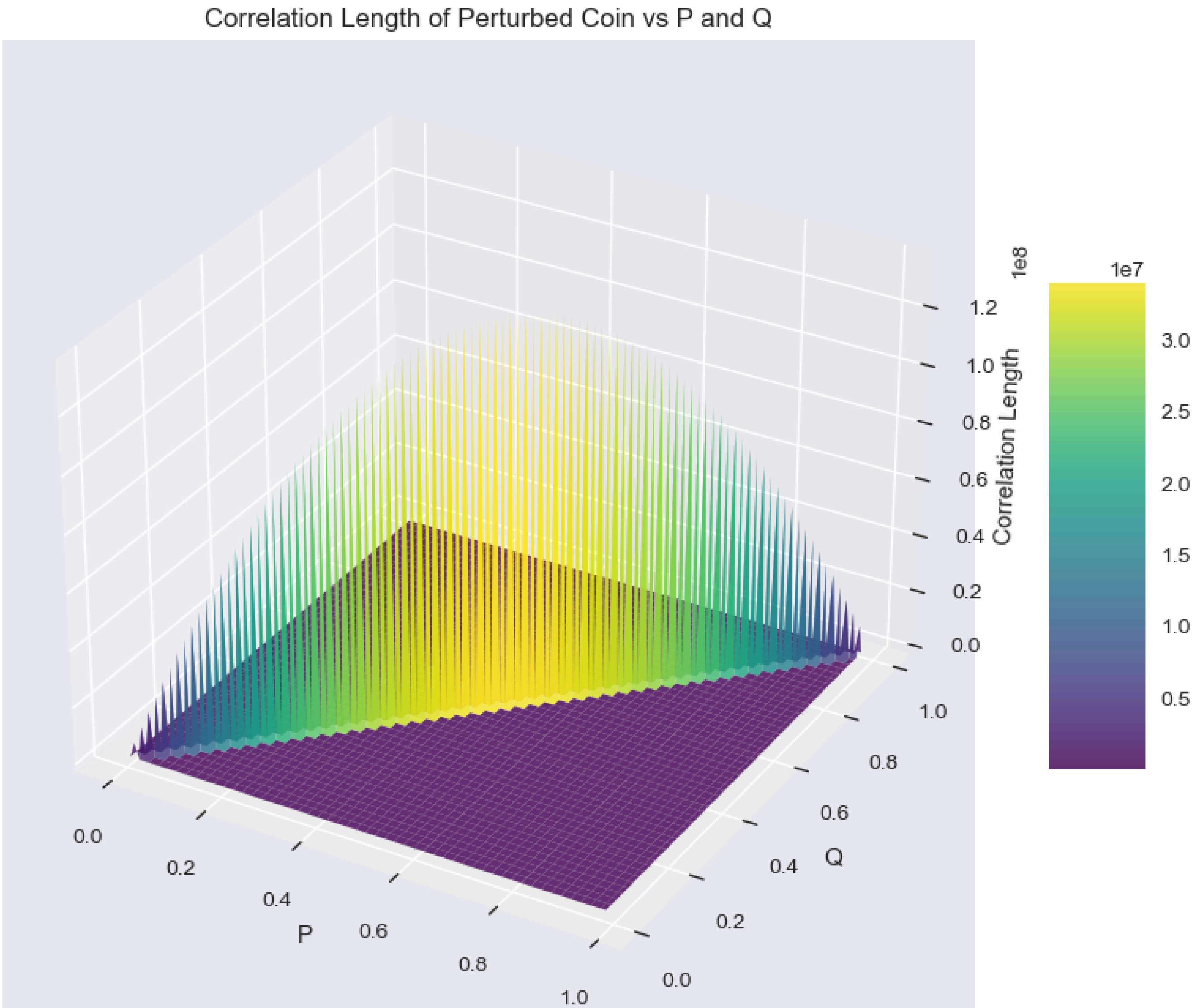
```
[[[0.6325 0.    ]
  [0.     0.4472]]

 [[0.     0.7746]
  [0.8944 0.    ]]]
```

Check Correlation Length

The system becomes memoryless when it approaches the line $p = q$, we can observe this with the plots below.

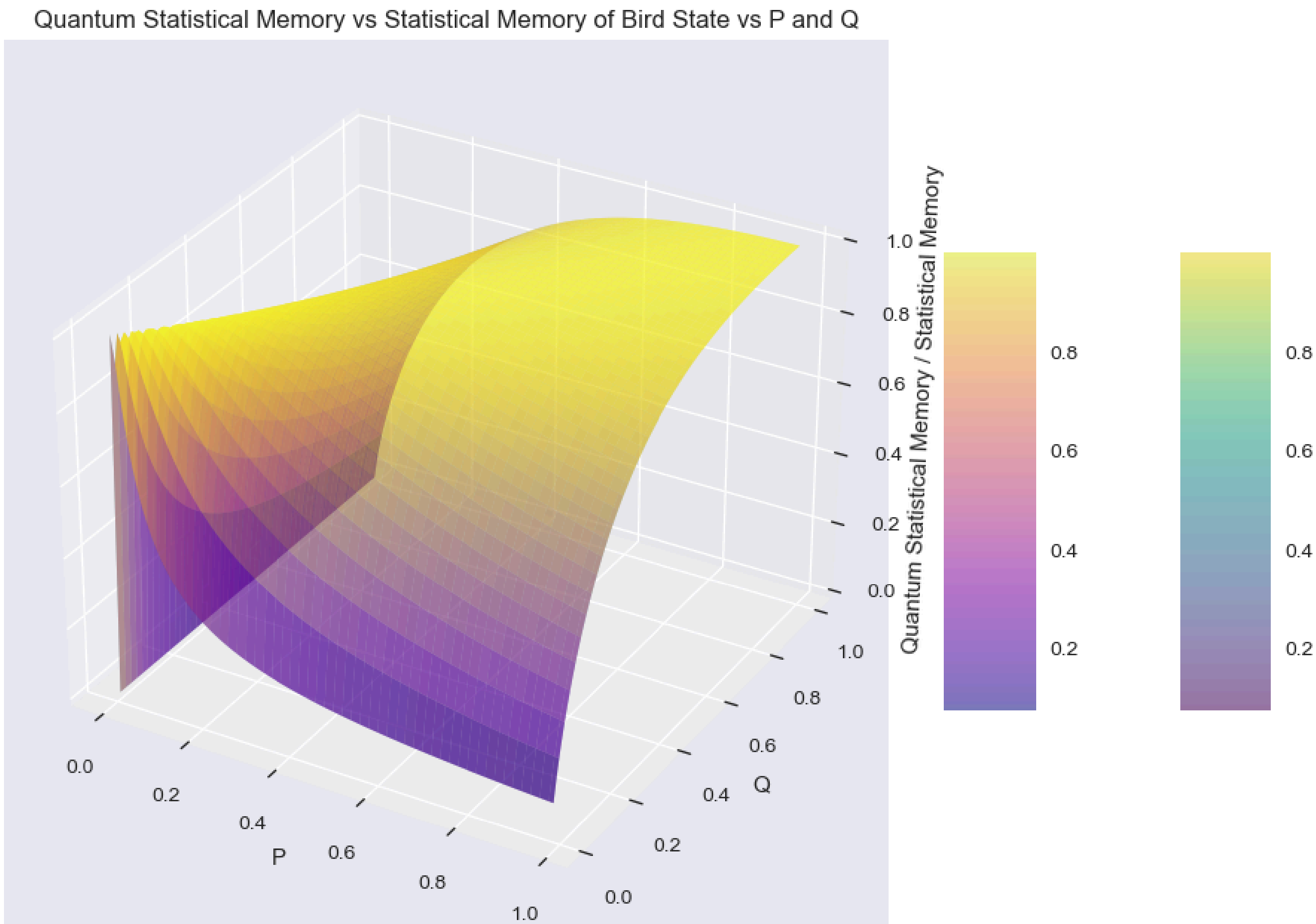


As we can see, the correlation length has a singularity when $p = q$.



The singularity is also shown here.

```
/opt/anaconda3/envs/physics_3_11_9/lib/python3.11/site-packages/scipy/linalg/_matfunc
s.py:219: LogmExactlySingularWarning: The logm input matrix is exactly singular.
  F = scipy.linalg._matfuncs_inv_ssq._logm(A)
```



Interestingly enough, the classical and quantum staistical memory overlaps with each other, hinting toward that there are no quantum advantage in modelling this system in the quantum manner??

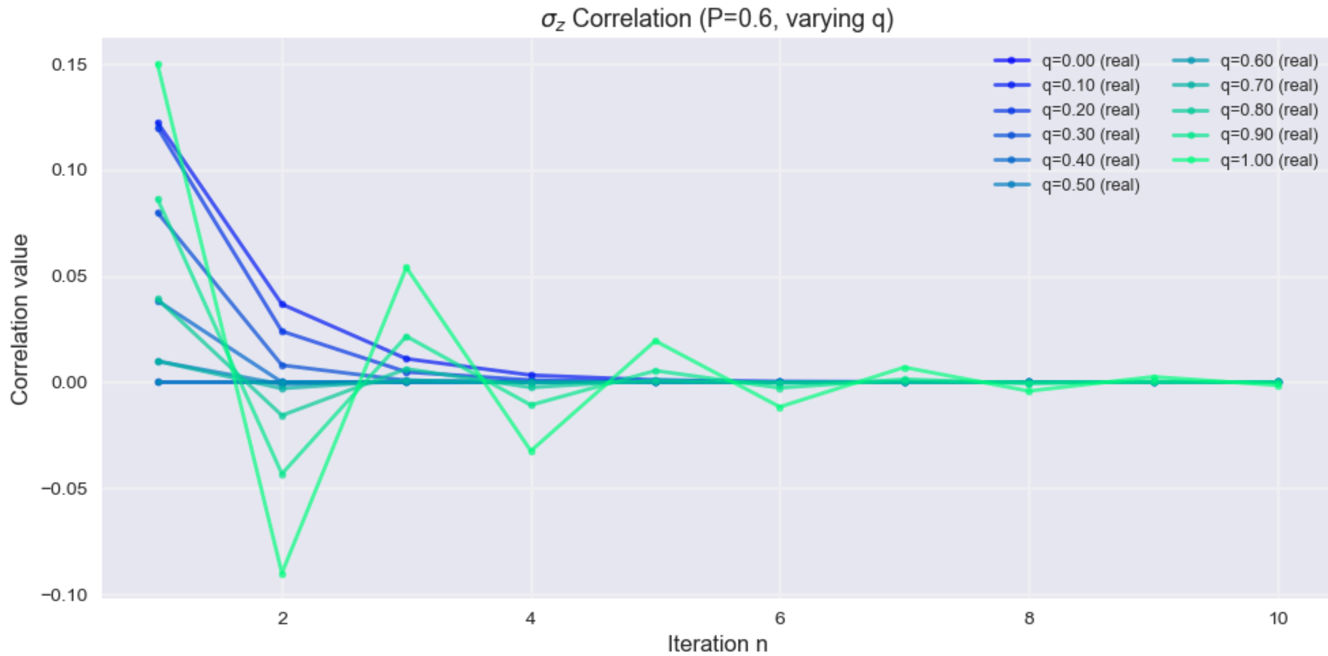
Correlation Function in the Bird Model

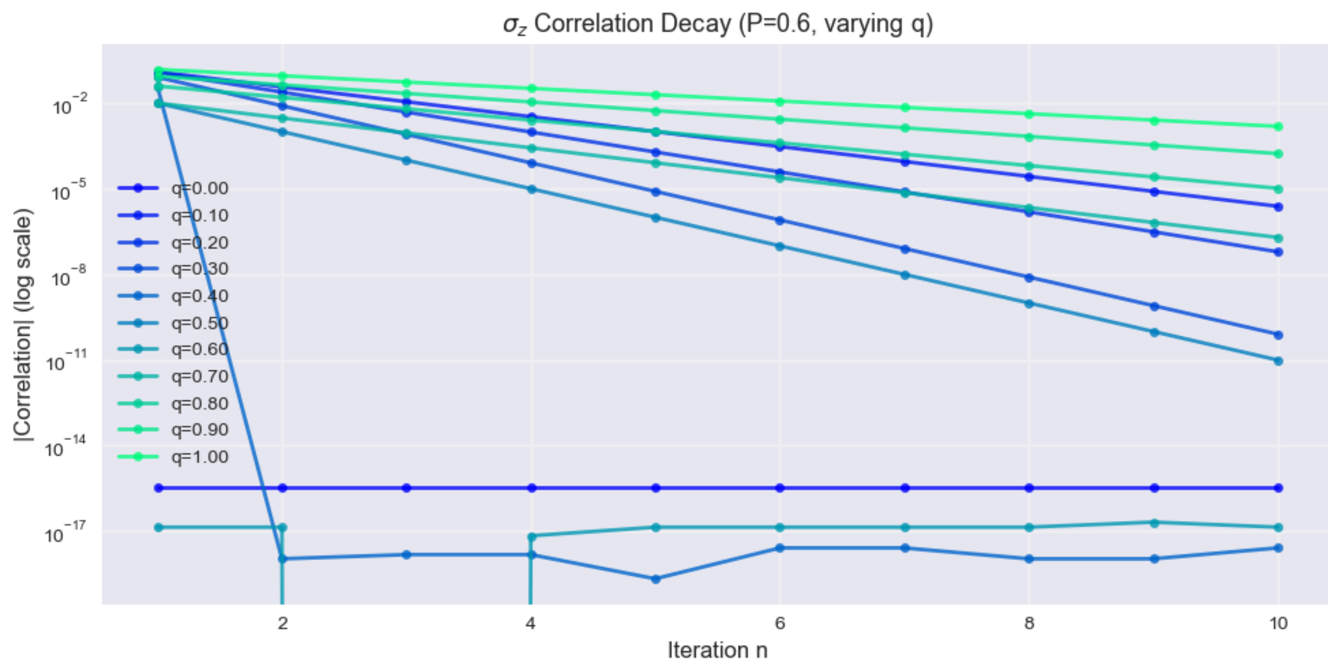
This section attempts to find the correlation behaviours of the Bird model with different operators.

Pauli σ_z Correlation Analysis

Analysis of correlation functions using the Pauli Z observable

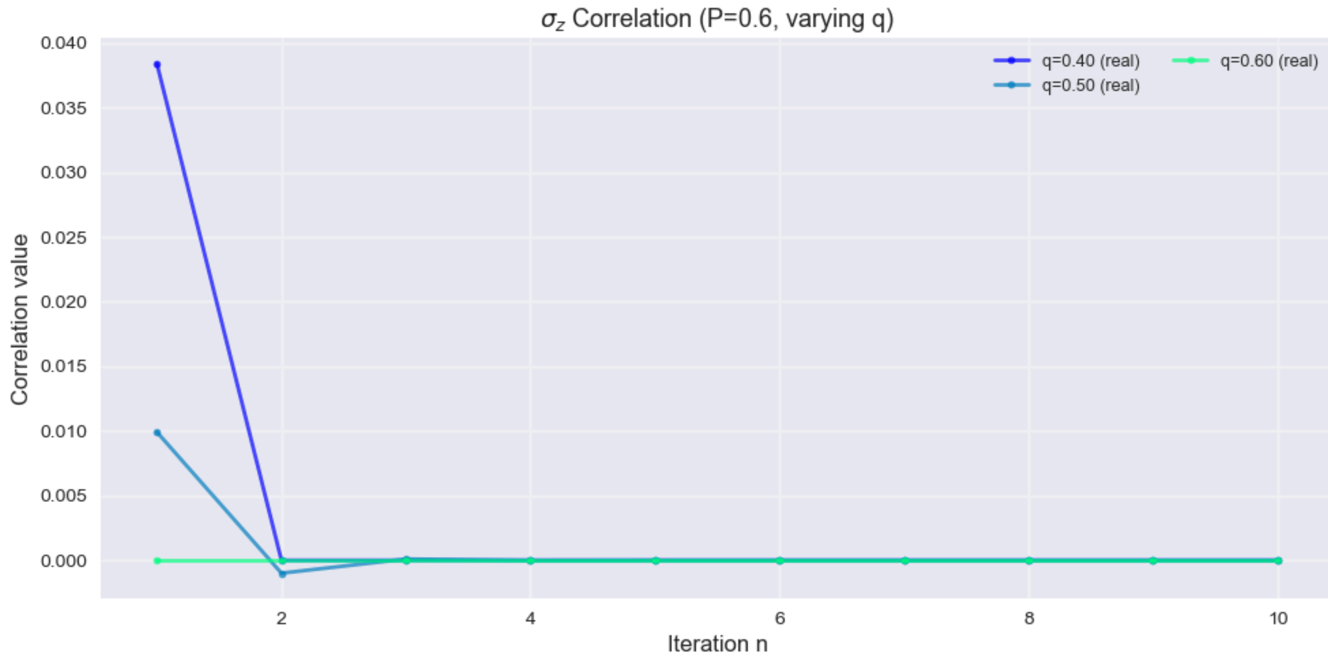
$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

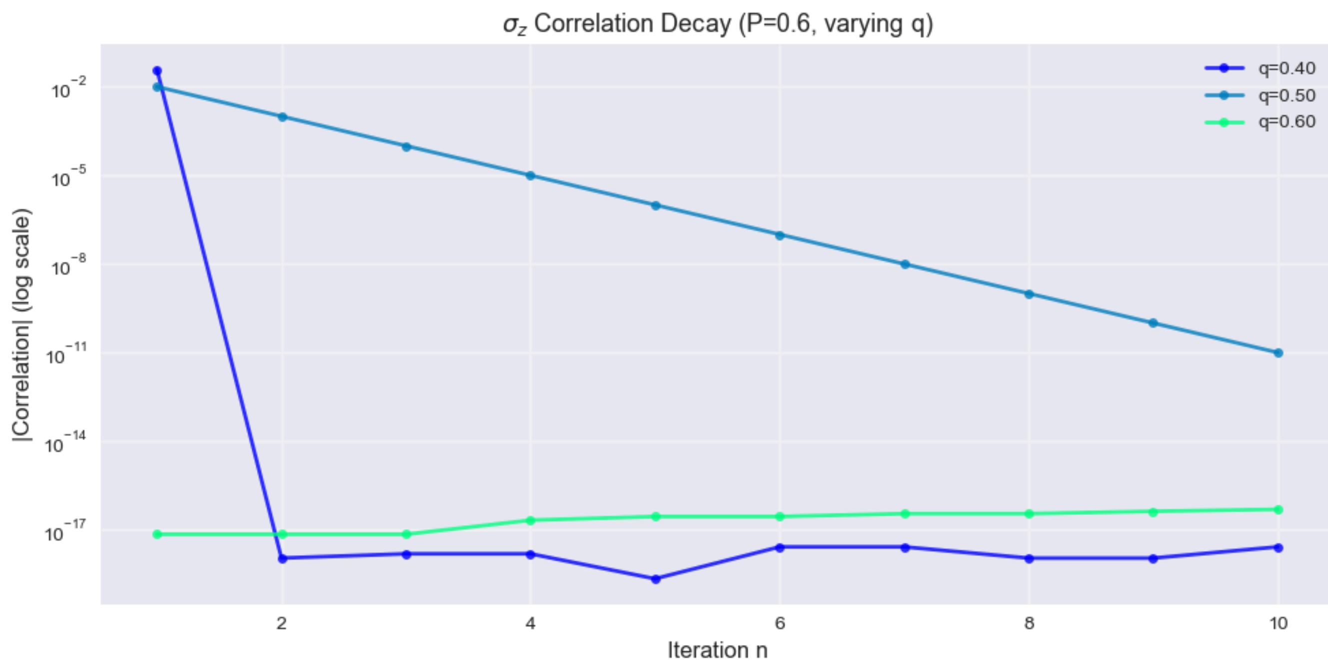





```
$\sigma_z$ Correlation Analysis with w_bird (fixed P=0.6, varying q)
q values tested: [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
```

Isolating the Weird Cases





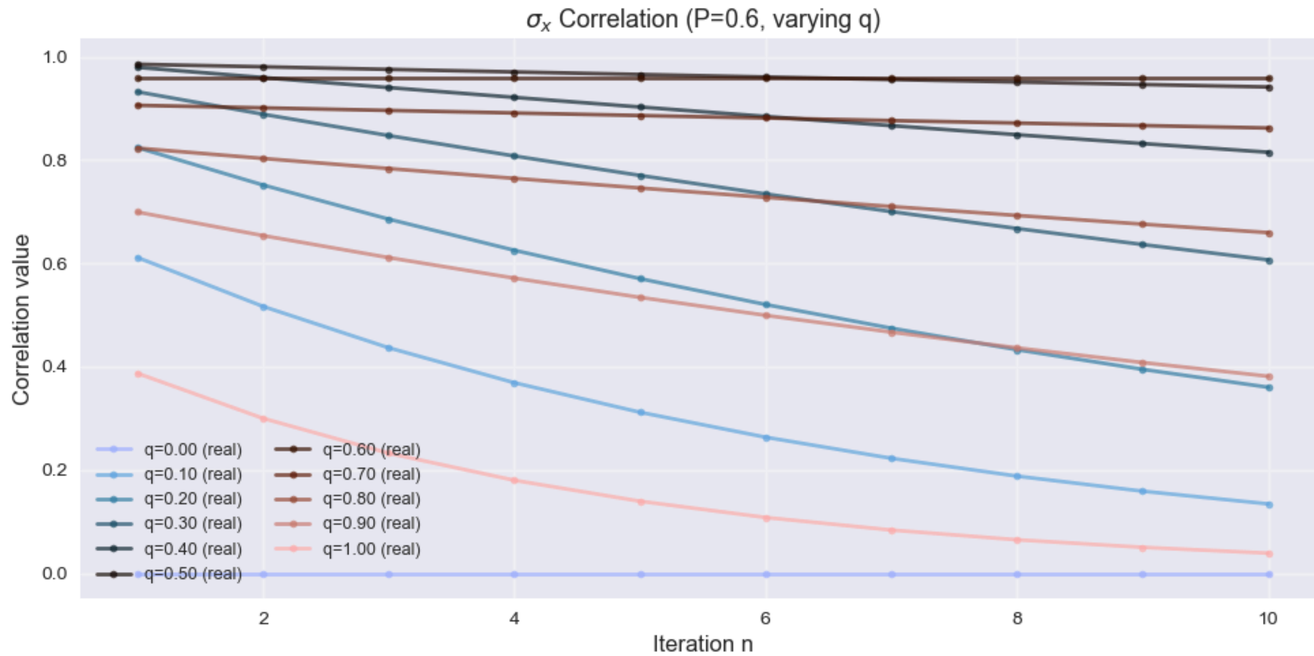

```
$\sigma_z$ Correlation Analysis with w_bird (fixed P=0.6, varying q)
q values tested: [0.4, 0.5, 0.6]
```

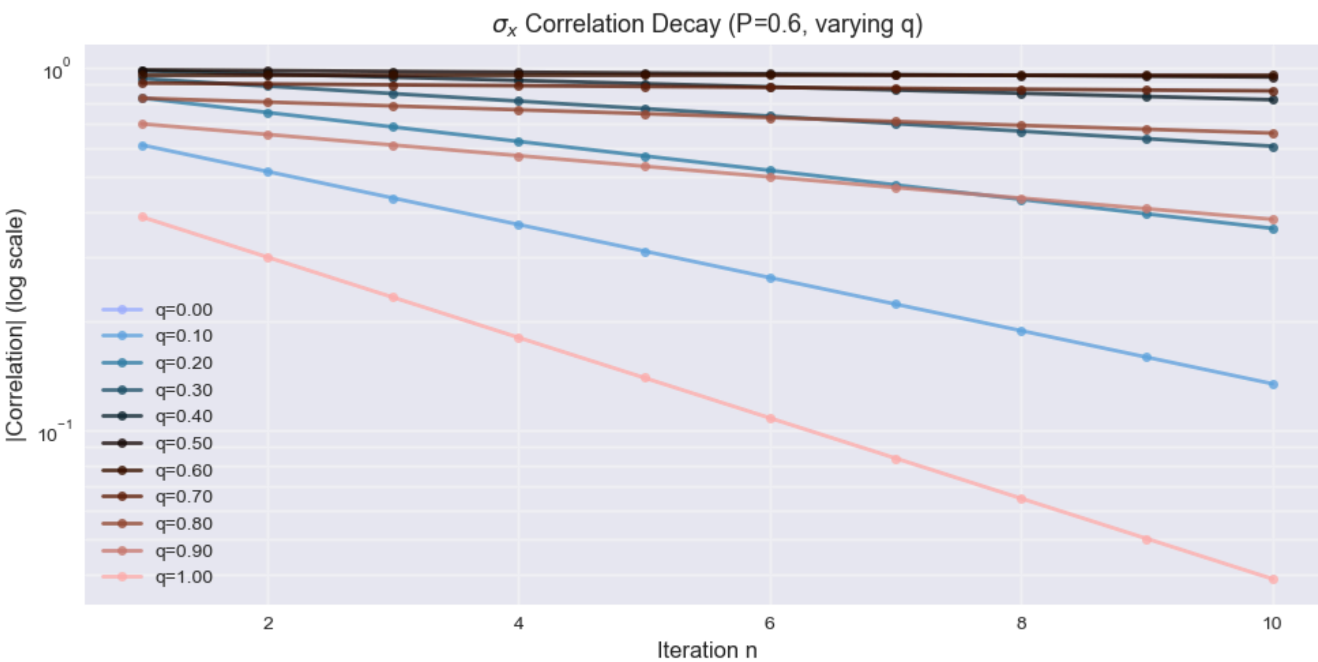
As we can see, though for the $q = 0.4$ doesn't seem that much like a singularity, its correlation in the z-basis drops extremely quickly. This probably have to do with the form of the correlation function when spectral decomposed (Ref. to my paper lol). Everything just seems to go back to normal for $q = 0.5$, and $q = 0.6$ is just dead.

Pauli σ_x Correlation Analysis

Analysis of correlation functions using the Pauli X observable

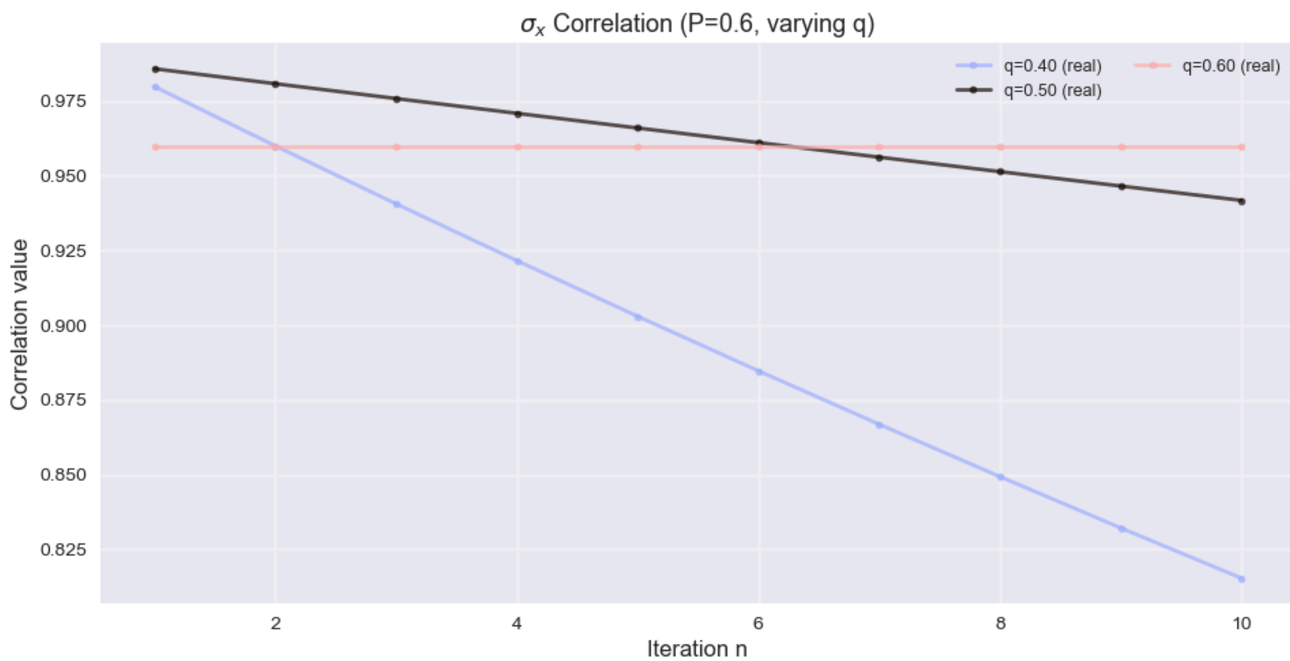
$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

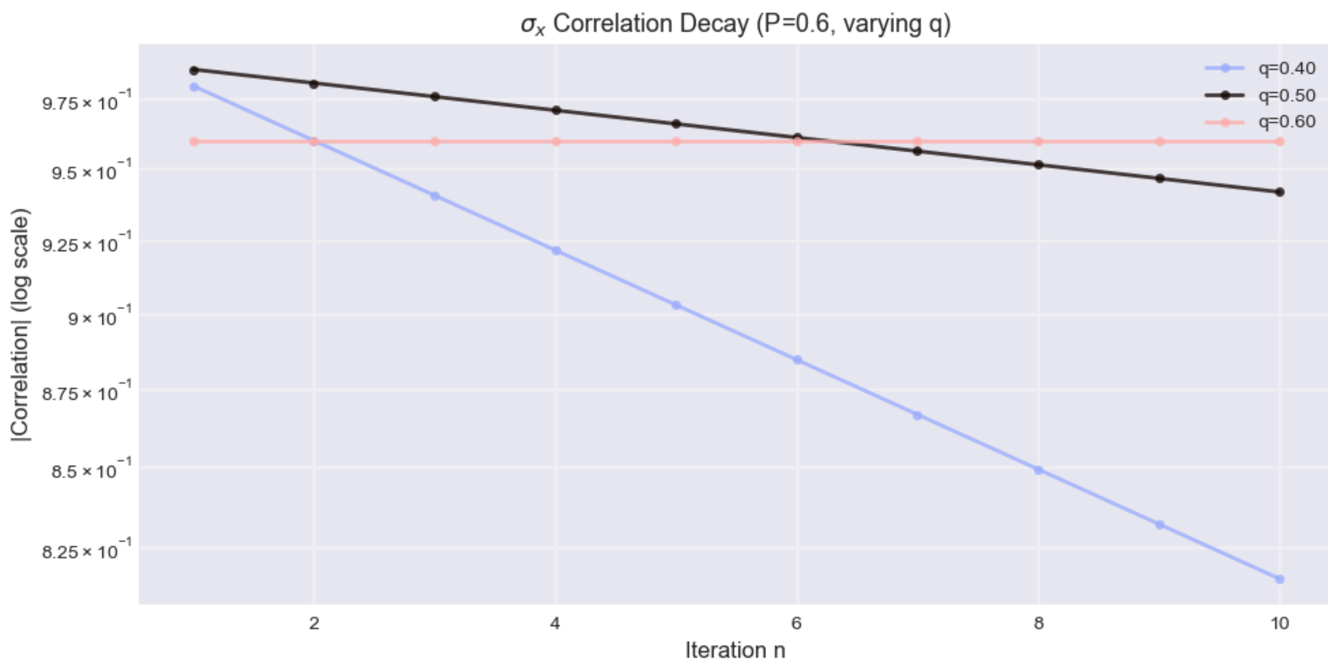





```
$\sigma_x$ Correlation Analysis with w_bird (fixed P=0.6, varying q)
q values tested: [0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
```

Again isolating cases around singularity.



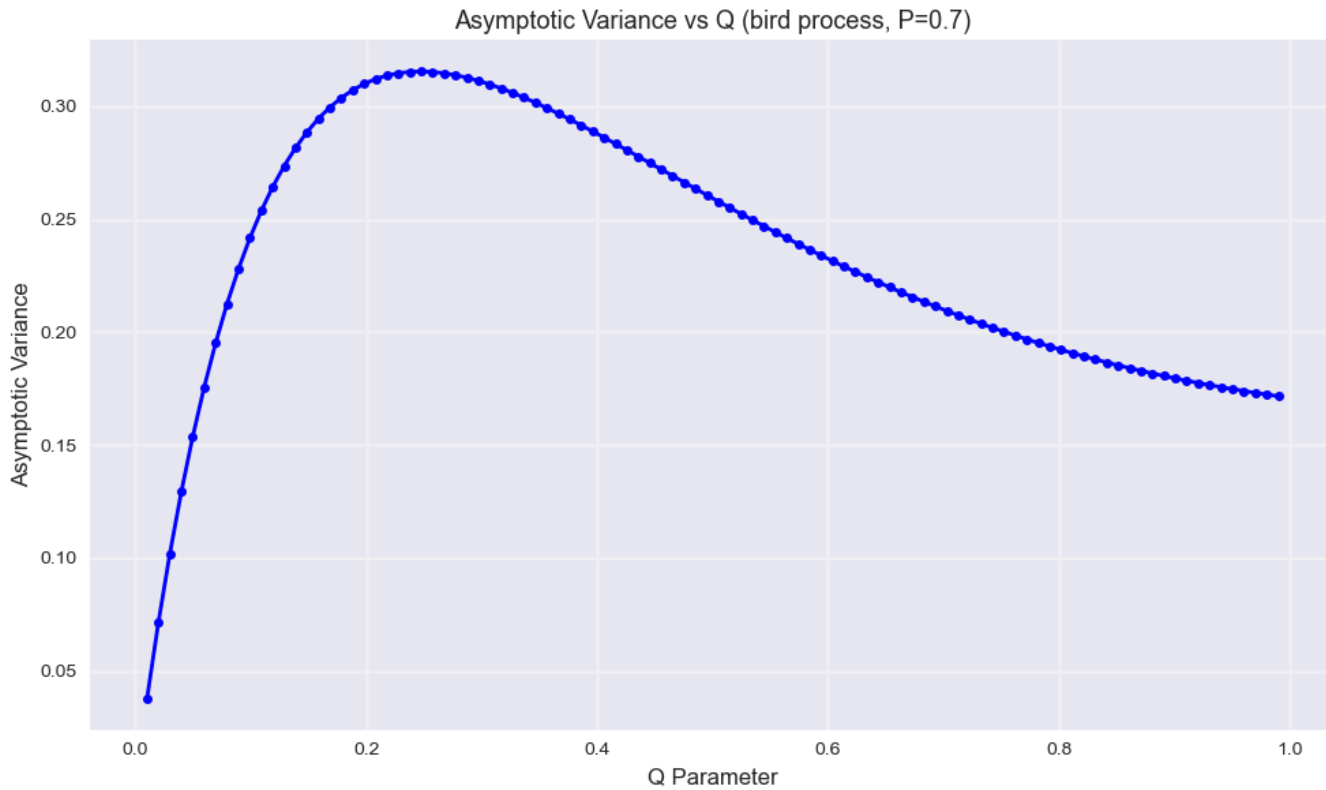



```
$\sigma_x$ Correlation Analysis with bird (fixed P=0.6, varying q)
q values tested: [0.4, 0.5, 0.6]
```

The behaviour in the x-basis looks much nicer, where we can see that there are no weird behaviour at $q = 0.4$ in this case, where instead of having no correlation as in z-basis, the correltaion in the x-basis when $q = 0.6$ persists forever and remains constant.

Asymptotic Variance Analysis (Bird process)

Analysis of asymptotic variance across varying Q parameter with fixed P for the w_bird process.




```
Asymptotic variance analysis for bird(P=0.7, Q)
Q range: 0.01 to 0.99
Variance range: 0.0380 to 0.3154
```

Symmetry Finding for the Bird

We now attempt to find the symmetry group generators for the perturbed coin process for a fixed value of $p \neq q$.

```
The parent hamiltonian in consideration is:
[[ 1.000e-04  0.000e+00  0.000e+00  4.100e-03  0.000e+00  4.100e-03
  -8.300e-03  0.000e+00]
 [ 0.000e+00  9.800e-03  4.820e-02  0.000e+00  4.820e-02  0.000e+00
   0.000e+00 -7.080e-02]
 [ 0.000e+00  4.820e-02  7.383e-01  0.000e+00 -2.617e-01  0.000e+00
   0.000e+00 -3.499e-01]
 [ 4.100e-03  0.000e+00  0.000e+00  6.648e-01  0.000e+00 -3.352e-01
  -3.324e-01  0.000e+00]
 [ 0.000e+00  4.820e-02 -2.617e-01  0.000e+00  7.383e-01  0.000e+00
   0.000e+00 -3.499e-01]
 [ 4.100e-03  0.000e+00  0.000e+00 -3.352e-01  0.000e+00  6.648e-01
  -3.324e-01  0.000e+00]
 [-8.300e-03  0.000e+00  0.000e+00 -3.324e-01  0.000e+00 -3.324e-01
   6.703e-01  0.000e+00]
 [ 0.000e+00 -7.080e-02 -3.499e-01  0.000e+00 -3.499e-01  0.000e+00
   0.000e+00  5.137e-01]]

================================================================================
SYMMETRIC GENERATORS ANALYSIS
================================================================================

⚠️  No symmetry found: The null space is empty.
================================================================================
```

Now for the case $p = q = 0.5$,

```
The parent hamiltonian in consideration is:
[[ 0.25  0.    0.   -0.25  0.    0.25 -0.25  0.  ]
 [ 0.    0.25 -0.25  0.    0.25  0.    0.   -0.25]
 [ 0.   -0.25  0.75  0.   -0.25  0.    0.   -0.25]
 [-0.25  0.    0.    0.75  0.   -0.25 -0.25  0.  ]
 [ 0.    0.25 -0.25  0.    0.25  0.    0.   -0.25]
 [ 0.25  0.    0.   -0.25  0.    0.25 -0.25  0.  ]
 [-0.25  0.    0.   -0.25  0.   -0.25  0.75  0.  ]
 [ 0.   -0.25 -0.25  0.   -0.25  0.    0.    0.75]]
Verifying that the MPS is in the ground space of the parent hamiltonian:
ph_3_3 @ state =  [0. 0. 0. 0. 0. 0. 0. 0.]

================================================================================
SYMMETRIC GENERATORS ANALYSIS
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⚠️  No symmetry found: The null space is empty.
================================================================================
```

Clearly there are no physical symmetry of the parent hamiltoniannfor this model.